

JAYOUNG KIM

Yuseong-gu ◇ Daejeon, South Korea

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RESEARCH INTERESTS

My research focuses on tabular data, spanning generation, representation learning, and foundation models. Recently, I have been exploring in-context learning approaches for tabular data generation. I am also broadly interested in data privacy and adversarial robustness.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Sep. 2024 - Present

Doctor of Philosophy in School of Computing (Adviser: Prof. Noseong Park)

Yonsei University

Mar. 2021 - Aug. 2024

Master of Science in Artificial Intelligence

Adviser: Prof. Noseong Park

Thesis: “STaSy: Score-based Tabular data Synthesis ”

Overall GPA: 4.19 / 4.3

Sookmyung Women’s University

Mar. 2016 - Feb. 2021

Bachelor of Science in Computer Science and Statistics

Overall GPA: 3.97 / 4.5

PUBLICATIONS

* denotes equal contribution.

Jayoung Kim, Kookjin Lee, Noseong Park, and Sanghyun Hong, “Adversarial Robustness of Implicit Neural Representation-Based Classifiers”, In *International Conference on Machine Learning (ICML)*, 2026. (*Under Review*)

Jayoung Kim, Yehjin Shin, and Noseong Park, “Polynomial-based Self-Attention for Table Representation Learning”, In *International Conference on Machine Learning (ICML)*, 2024.

Chaejeong Lee*, **Jayoung Kim***, and Noseong Park, “CoDi: Co-evolving Contrastive Diffusion Models for Mixed-type Tabular Synthesis”, In *International Conference on Machine Learning (ICML)*, 2023.

Jayoung Kim*, Chaejeong Lee*, and Noseong Park, “STaSy: Score-based Tabular data Synthesis”, In *International Conference on Learning Representations (ICLR)*, 2023. [**Spotlight: notable-top-25%**]

Jihyeon Hyeong, **Jayoung Kim**, Noseong Park, and Sushil Jajodia, “An Empirical Study on the Membership Inference Attack against Tabular Data Synthesis Models”, In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM)*, 2022.

Jaehoon Lee, Jinsung Jeon, Sheo yon Jhin, Jihyeon Hyeong, **Jayoung Kim**, Minju Jo, Seungji Kook, and Noseong Park, “LORD: Lower-Dimensional Embedding of Log-Signature in Neural Rough Differential Equations”, In *International Conference on Learning Representations (ICLR)*, 2022.

Sheo Yon Jhin, Jaehoon Lee, Minju Jo, Seungji Kook, Jinsung Jeon, Jihyeon Hyeong, **Jayoung Kim** and Noseong Park, “EXIT: Extrapolation and Interpolation-based Neural Controlled Differential Equations for Time-series Classification and Forecasting”, In *Proceedings of the Web Conference (WWW)*, 2022.

Jayoung Kim*, Chaejeong Lee*, Yehjin Shin, Sewon Park, Minjung Kim, Noseong Park, and Jihoon Cho, “SOS: Score-based Oversampling for Tabular Data”, In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD)*, 2021.

Jayoung Kim*, Jinsung Jeon*, Jaehoon Lee, Jihyeon Hyeong, and Noseong Park, “OCT-GAN: Neural ODE-based Conditional Tabular GANs”, In *Proceedings of the Web Conference (WWW)*, 2021.

Jinsung Jeon, Jing Lu, **Jayoung Kim**, Jaehoon Lee, Noseong Park, Jamie Jooyeon Lee, Ozlem Uzuner, and Sushil Jajodia, “Scalable Graph Synthesis with Adj and 1 - Adj”, In *Proceedings of the 2021 SIAM International Conference on Data Mining (SDM)*, 2021.

Patents

[Granted] Neural Ordinary Differential Equations-based Conditional Tabular Data Adversarial Generative Neural Network Apparatus and Method, Noseong Park, **Jayoung Kim**, Jinsung Jeon, Jaehoon Lee, Jihyeon Hyeong, Korean Patent (Registration No. 10-2839821), Registered: July 2025.

Also filed in the United States (Application No. 17/564,870) and Japan (Application No. 2021-215113).

[Filed] Time-series Analysis Method and Apparatus using Extrapolation and Interpolation-based Neural Controlled Differential Equations, Noseong Park, Sheoyon Jhin, Jaehoon Lee, Minju Jo, Seoungji Kook, Jinsung Jeon, **Jayoung Kim**, Korean Patent Application (Application No. 10-2022-0180315), Filed: December 2022.

PROJECTS

Approximate Query DBMS for Big Data Exploratory Analysis 2021 - 2024

- Collaborative research project with ETRI (Electronics and Telecommunications Research Institute)
- Developed synopsis structures using table generative models to enable fast query processing on large-scale data
- Designed continual learning-based methods to update synopses reflecting changes in raw data efficiently
- Resulted in publications at ICML, ICLR, and CIKM

Collaborative Research with Samsung SDS 2021 – 2023

- Proposed generative models for table synthesis and oversampling strategies
- Resulted in a publication at KDD 2021

INVITED TALKS

Tutorial on Graph Neural Networks and Generative Models, KSIAM Winter AI Tutorial, 2023

Invited Talk on Tabular Data Generation, Data Systems Lab, POSTECH, 2023

AWARDS

Excellent Paper Award in the 1st Yonsei AI workshop with “SOS: Score-based Oversampling for Tabular Data” 2022

ACADEMIC SERVICE

Reviewer

ICML (2026), ICLR (2024–2026), NeurIPS (2023–2025), TMLR (2025)

Teaching Assistant

Machine Learning (Spring 2025, Spring 2026), Deep Learning Design Theory (Fall 2024), Algorithm Analysis (Fall 2021)

SKILLS

Programming Languages Python

Deep Learning PyTorch, Mixed Precision Training, Multi-GPU Training

Tools Git, Weights & Biases, Linux